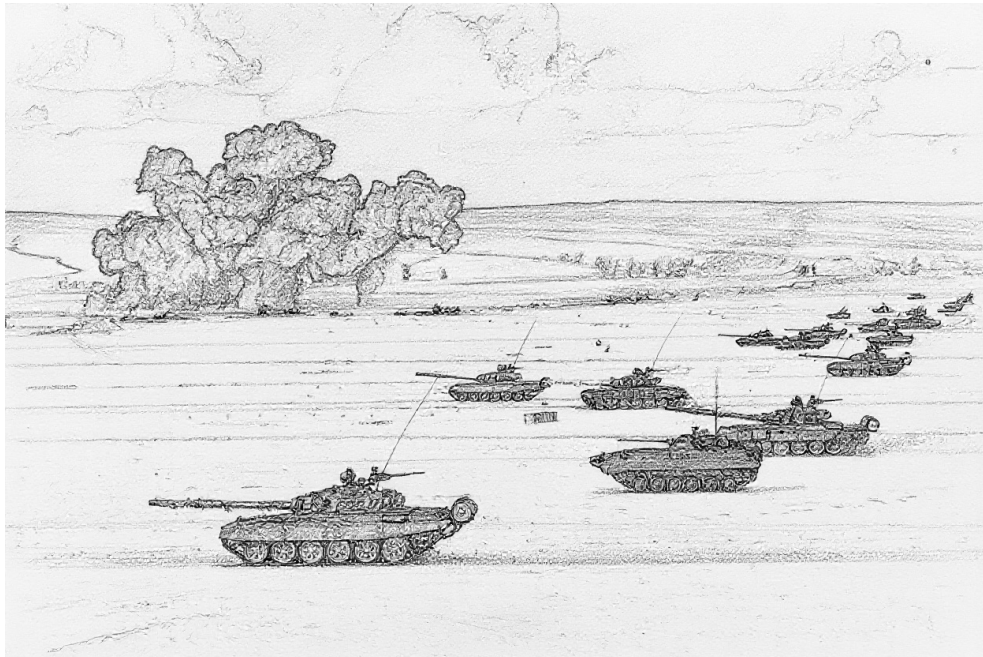


Air Power

Ground Unit Briefing

Alan Watson Forster

17 June 2026



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PREFACE

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:¹

“You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb.”

This briefing document deals with the ground units and ground targets that are subject to air-to-ground attacks. It first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from J.D. Webster’s *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from Malcolm Pipe’s extensive compendium of AAA and SAM units, and to a lesser degree from *Air Power Journal*. However, the data have been carefully revised for accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded. Extensive notes follow the main text, providing provenance of data, sources of information, and justifications for decisions.

I am grateful to Carl Smeaton for many discussions on the data presented here and to Malcolm Pipes for leading the way with his compilations of AAA and SAM units.

Alan Watson-Forster
23 June 2026

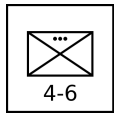
GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide operational and strategic mobility to infantry and towed artillery and are the lifeblood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. Some representative examples of vehicles are given in the descriptions, and the Vehicle Types Table gives a more extensive list. All ground-combat and transport platoons and batteries are also available as sections and squads.²

INFANTRY UNITS

Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

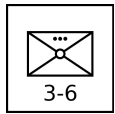
An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.



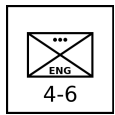
Infantry Platoon: An infantry platoon is armed with small arms, light machine guns, and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. The platoon is the primary combat unit in larger infantry formations.



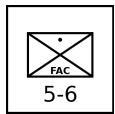
Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation.³



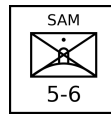
Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is often a battalion-level asset, and its role is to provide indirect fire support and smokescreens to the infantry platoons in its formation. Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even in light infantry formations.



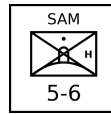
Infantry Engineers Platoon: An infantry engineers platoon is additionally equipped and trained for combat engineering tasks such as mine-laying and mine-clearing. It is often battalion-level asset.



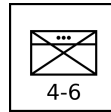
Infantry FAC Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection. An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere. An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section. An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

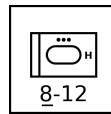


Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.⁴

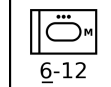
ARMORED VEHICLE UNITS

Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a fire-power or transport capability.

The primary factors that determine an armored-vehicle unit's type are its weapons and protection level. The means of traction is a secondary consideration. Armored cars and armored half-tracks are considered to be light tanks, light IFVs, or light or open APCs, light anti-tank missile vehicles, or light reconnaissance vehicles according to their role and armament.



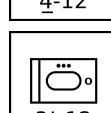
Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns such as the SU-100, Kanonen JPz, and Strv 103 are also considered to be tanks. The role of a tank is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor.⁵



Heavy examples: M1 Abrams, T-64, and T-72/80.

Medium examples: M60, T-34, T-54/55, and T-62.

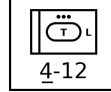
Light examples: M114A2, M551 Sheridan, and PT-76.



Open example: ASU-57.

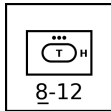


IFV Platoon: An infantry fighting vehicle (IFV) platoon is equipped with armored vehicles that provide infantry with tactical and operational mobility, protection, and fire support, being armed with a 20 mm or larger cannon and sometimes an anti-tank missile launcher. In most cases, the infantry must dismount to fight effectively. An IFV platoon may have medium or light armor.⁶



Medium example: M2A2 Bradley.

Light examples: M2/M2A1 Bradley and BMP-1/2.



8-12

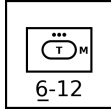
APC Platoon: An APC platoon is equipped with armored vehicles that provide infantry with tactical and operational transport and protection, but unlike IFVs do not provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor. Heavy and medium APCs are typically converted tanks.⁷

Heavy example: Nakpadon.

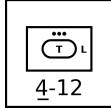
Medium example: Nagmachon.

Light examples: M113 and BTR-50PK/60PB/70.

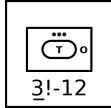
Open examples: M3 half-track and BTR-50P/60P.



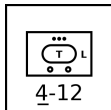
6-12



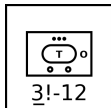
4-12



3!-12



4-12

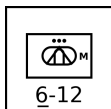


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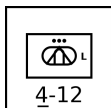
Armored Truck Platoon: An armored truck platoon has armored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. Unlike a truck platoon, it has armored protection, but unlike an APC platoon, it has more limited off-road mobility. An armored truck platoon may have light or open armor.⁸

Light example: BTR-152K.

Open examples: BTR-40 and other BTR-152 versions.



6-12

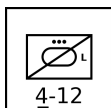


4-12

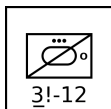
Armored Anti-Tank Missile Platoon: An armored anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers such as the Kanonen JPz are considered to be tanks. An armored anti-tank missile platoon is often a battalion- or regiment-level asset. It may have medium or light armor.⁹

Medium examples: Raketen JPz and Jaguar.

Light examples: M150, M901, and BRDM.



4-12

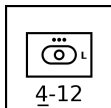


3!-12

Armored Reconnaissance Platoon: An armored reconnaissance platoon has armored vehicles specialized in scouting and observation and with no offensive weapons more powerful than machine guns. (Vehicles used for reconnaissance that do have significant offensive weapons are typically classified as light tanks or IFVs.) An armored reconnaissance platoon may have light or open armor.¹⁰

Light examples: M114 and BRDM-2.

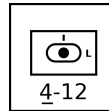
Open example: M20.



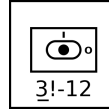
4-12

Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (most commonly converted APCs or IFVs). It is typically a battalion-level asset in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor.¹¹

Examples: M106 and M125.



4-12

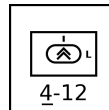


3!-12

Armored Artillery Battery: An armored artillery battery has guns or howitzers for indirect-fire support in armored vehicles providing some protection, in particular from counter-battery fire, and mobility. It is typically a regiment-, brigade-, or division-level asset. An armored artillery battery may have light or open armor.¹²

Light examples: M109, 2S3, and 2S1.

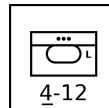
Open example: M7 Priest.



4-12

Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor.¹³

Example: M270.

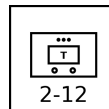


4-12

Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

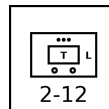
UNARMORED VEHICLE UNITS

Unarmored or soft vehicles provide mobility, but do not provide (or provide only incomplete) armor protection for their crew and weapons. They include armored vehicles whose crew or weapons are exposed. They may be wheeled or tracked.



2-12

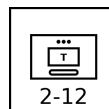
Truck Platoon: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. It can provide operational mobility to infantry and towed artillery, but due to its extreme vulnerability to direct- and indirect-fire, is not typically used in a tactical role.



2-12

Light Truck Platoon: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies. It can provide transport to independent infantry FAC or SAM units, but in these roles it can avoid close contact with enemy combat units, mitigating its vulnerability.

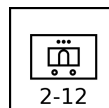
Examples: M151 Jeep, HMMWV, and GAZ-69.



2-12

Tracked Carrier Platoon: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition. It serves the same roles as trucks or light trucks, but in terrain that requires their greater mobility, in particular arctic regions.

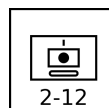
Example: Bv 202.



2-12

Mobile Anti-Tank Missile Platoon: A mobile anti-tank missile platoon is equipped with anti-tank missiles mounted on unarmored light trucks. It is extremely vulnerable to direct- and indirect-fire, and must rely on concealment and mobility.

Examples: M151 TOW and M966 HMMWV TOW.

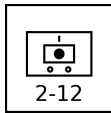


2-12

Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its

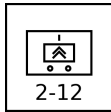
role is indirect fire support.¹⁴

Examples: M107, M110, 2S4, and 2S7.



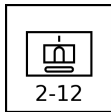
Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. Lacking armor, it must avoid counter-battery fire by using their mobility to rapidly change firing positions. It is typically a regiment-, brigade-, or division-level asset, and their role is indirect fire support.¹⁵

Examples: CAESAR and Archer.



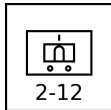
Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery. Again, it relies on mobility to avoid counter-battery fire.¹⁶

Examples: M142 HIMARS and BM-21 Grad.



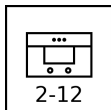
Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a tracked artillery battery. The tracked chassis provides mobility, but the missiles are exposed. Nevertheless, the long range of the missile permits deployment far behind the front line, mitigating its vulnerability.¹⁷

Example: M752 Lance.



Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery. Again, the long range of the missile permits deployment far behind the front line, mitigating its vulnerability.¹⁸

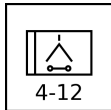
Examples: M1003 Pershing II and OTR-21 Tocha (SS-21).



Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

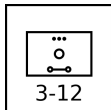
TOWED UNITS

Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to similar vehicle-mounted weapons. Towed weapons are typically provided with appropriate ground or helicopter transport.



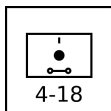
Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Example: 2A18 (T-12) and 2A45 (Sprut).



Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons.

Example: M1943.



Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Examples: M119, M198, and 2A18 (D-30).

GROUND-COMBAT AND TRANSPORT UNIT TABLE

Type	Size ^a	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class ^b
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	-/-/8	—
Infantry SAM	Squad	5	6	U	—	—
Infantry Heavy SAM	Squad	5	6	U	—	—
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Heavy Tank	Platoon	8	12	U	12/8/4	B3
Medium Tank	Platoon	6	12	U	10/7/3	B3
Light Tank	Platoon	4	12	U	6/4/2	B3
Open Tank	Platoon	3!	12	U	5/3/2	B3
Medium IFV	Platoon	6	12	U	10/7/3	B3
Light IFV	Platoon	4	12	U	6/4/2	B3
Heavy APC	Platoon	8	12	U	10/7/3	B3
Medium APC	Platoon	6	12	U	8/5/3	B3
Light APC	Platoon	4	12	U	6/4/2	B3
Open APC	Platoon	3!	12	U	5/3/2	B3
Light Armored Truck	Platoon	4	12	G/P	6/4/2	B3
Open Armored Truck	Platoon	3!	12	G/P	5/3/2	B3
Medium Armored Anti-Tank Missile	Platoon	6	12	U	12/8/4	B3
Light Armored Anti-Tank Missile	Platoon	4	12	U	10/7/3	B3
Light Armored Reconnaissance	Platoon	4	12	U	6/4/2	B3
Open Armored Reconnaissance	Platoon	3!	12	U	4/3/1	B3
Light Armored Mortar	Platoon	4	12	U	10/7/3	B3
Light Armored Artillery	Battery	4	12	U	12/8/4	B3
Open Armored Artillery	Battery	3!	12	U	12/8/4	B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4	B3
Light Armored HQ	Platoon	4	12	U	12/8/4	B3
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—
Mobile Anti-Tank Missile	Platoon	2	12	G/P	8/5/3	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Towed Mortar	Platoon	3	12	T	7/5/2	—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—

^a Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs. ^b Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

VEHICLE TYPES TABLE

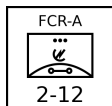
Heavy Tanks		Light Tanks		Heavy APCs		Medium Armored Anti-Tank		Light Truck	
Al-Khalid/VT-1A	PK/CN 2001	AMX-10 RC	FR 1981	Achzarit	IL 1988	Raketen JPz 1	DE 1961	GAZ-69	SU 1953
Ariete	IT 1995	AMX-13	FR 1953	Nakpadon	IL 1998	Raketen JPz 2	DE 1967	HMMWV	US 1985
Arjun	IN 2004	Fox	GB 1975	Namer	IL 2008	Jaguar 1	DE 1978	Ilitis	GE 1978
Challenger 1	GB 1983	Commando (90 mm)	US 1964	Puma	IL 1993	Jaguar 2	DE 1983	Landrover	UK 1949
Challenger 2	GB 1998	LAV I Cougar	CA 1976	Medium APCs		Light Armored Anti-Tank		M38 1/4-ton truck	US 1949
Chieftain	GB 1967	LAV II Coyote	CA 1990	Boxer APC	NL/DE 2011	BRDM-2 ATGM	SU 1973	M151 1/4-ton truck	US 1960
Conqueror	GB 1955	M8/Greyhound	US 1943	Nagmachon	IL 2000	BRDM-3 ATGM	SU 1973	UAZ-469	SU 1971
IS-2	SU 1944	M24 Chaffee	US 1944	Nagmashot	IL 1990	Fennek MRAT	DE/NL 2003	Willys MB 1/4-ton truck	US 1941
IS-3	SU 1945	M41 Walker Bulldog	US 1953	Light APCs		FV438 Swingfire	GB 1971	Tracked Carrier	
K1	KR 1987	M113 C&V with 25 mm	US 1974	Bronco/Warthog	SG 2001	M150	US 1973	AT-T/S/L	SU 1950
K2 Black Panther	KR 2014	M114A2	US 1969	BvS10/Viking	SE 2005	M901 ITV	US 1979	Bv 202	SE 1964
Leclerc	FR 1992	M551 Sheridan	US 1969	BTR-50PK	SU 1958	Striker	GB 1976	Bv 206	SE 1980
Leopard 2	DE 1979	Panhard AML	FR 1961	BTR-60PA/PAI/PB	SU 1965	Stryker ATGM	US/CA 2003	BvS10/Beowulf	SE 2024
M-84	YU 1985	Panhard EBR	FR 1951	BTR-70	SU 1972	VAB HOT	FR 1978	M548	US 1960
M1 Abrams	US 1980	PT-76	SU 1951	BTR-80	SU 1986	YP-408 PRAT	NL 1976	Sisu Nasu	FI 1986
Merkava 1	IL 1979	Saladin	GB 1958	Commando	US 1963	YPR-765 PRAT	NL 1980	Mobile Anti-Tank Missile	
Merkava 2	IL 1983	Scimitar	GB 1973	FV432	GB 1963	Light Armored Reconnaissance		M151 TOW	US 1970
Merkava 3	IL 1989	Scorpion	GB 1973	LAV I Grizzly	CA 1976	BRDM-2	SU 1973	M966 HMMWV TOW	US 1985
Merkava 4	IL 2004	SpPz Luchs	DE 1975	LAV II Bison	CA 1990	Fennek	DE/NL 2003	Mobile Artillery	
PT-91	PL 1995	Stryker MGS	US/CA 2006	LVTP-5	US 1956	Ferret	GB 1952	Archer	SE 2016
Stridsvagn 103	SE 1967	Strv 74	SE 1958	LVTP-7 (AAV)	US 1972	M113 C&V/Lynx	US 1966	CAESAR	FR 2004
T-10	SU 1954	Type 62	CN 1961	M59	US 1954	M114	US 1962	Tracked Artillery	
T-14 Armata	RU 2024	Type 63	CN 1963	M75	US 1952	Open Armored Reconnaissance		2S4 Tyulpan	SU 1975
T-64	SU 1966	Type 05 AAV	CN 2005	M113	US 1960	M20	US 1943	2S7 Pion	SU 1976
T-72	SU 1973	Type 08 ARV	CN 2008	OT-64 SKOT	CS/PL 1963	Light Armored Mortar		M41	US 1942
T-80	SU 1976	Open Tanks		OT-810 half-track	CS 1958	FV432 with 81 mm	GB 1963	M107	US 1962
T-84	UA 1999	ASU-57	SU 1951	Piranha APC	CH 1974	M106	US 1964	M110	US 1963
T-90	RU 1992	SU-76	SU 1943	Saracen	GB 1952	M125	US 1964	AMX-13 F3	FR 1962
Type 10	JP 2012	Medium IFVs		Spartan	GB 1978	M1064	US 1990	Mobile Rocket Artillery	
Type 90	JP 1990	BMP-3	SU 1987	Stryker ICV	US/CA 2002	Stryker MCV	US/CA 2005	BM-8	SU 1941
Type 99	CN 2001	Boxer IFV	NL/DE 2011	TPz Fuchs	DE 1979	Light Armored Artillery		BM-13	SU 1941
Medium Tanks		Strf 9040/CV90	SE 1994	Type 63	CN 1978	2S1 Gvozдика	SU 1972	BM-31	SU 1944
AMX-30	FR 1966	LAV III (add-on-armor)	CA 2009	Type 77	CN 1978	2S3 Akatsiya	SU 1971	BM-24	SU 1947
ASU-85	SU 1959	M2A2/3/4 Bradley	US 1988	Type 89	CN 1978	2S19 Msta-S	SU 1989	BM-14	SU 1952
Centurion	GB 1945	M3A2/3/4 Bradley	US 1988	Type 08 APC	CN 2008	AMX-30 AuF1	FR 1977	BM-21 Grad	SU 1963
Comet	GB 1944	SPz Marder	DE 1971	VAB	FR 1979	AS-90	GB 1992	BM-27 Uragan	SU 1975
Cromwell	GB 1944	SPz Puma	DE 2015	YP-408	NL 1964	Bkan 1	SE 1967	BM-30 Smerch	SU 1989
ISU-122	SU 1944	Light IFVs		Open APCs		L-33 Ro'Em	IL 1973	M142 HIMARS	US 2005
ISU-152	SU 1943	AIFV	US 1979	AT-P	SU 1954	M52	US 1955	Tracked Missile Artillery	
Kanonen JPz	DE 1965	AMX-10P	FR 1973	BTR-50P	SU 1952	M108	US 1962	M752 Lance	US 1972
Leopard 1	DE 1965	BMP-1	SU 1966	BTR-60P	SU 1959	M109	US 1963	AMX-30 Pluton	FR 1974
M-50 Sherman	IL/FR 1956	BMP-2	SU 1980	M3 half-track	US 1941	Panzerhaubitze 2000	DE 1998	Mobile Missile Artillery	
M-51 Sherman	IL/FR 1961	Commando (20 mm)	US 1963	Light Armored Trucks		Open Armored Artillery		9K52 Luna-M (Frog-7)	SU 1962
M103	US 1957	FV432 RARDEN	GB 1975	BTR-152K	SU 1959	AMX-105A	FR 1958	Corporal	US 1954
M26 Pershing	US 1944	LAV III	CA 1999	Saxon	GB 1983	M-50	IL 1962	Sergeant	US 1962
M4 Sherman	US 1942	LAV-25	CA 1983	Open Armored Trucks		M7 Priest	US 1942	M34/M50 Honest John	US 1953
M46 Patton	US 1949	M2 and M2A1 Bradley	US 1981	BTR-40	SU 1950	M40	US 1945	M74 Redstone	US 1958
M47 Patton	US 1951	M3 and M3A1 Bradley	US 1981	BTR-152A	SU 1950	M44	US 1954	M790 Pershing 1a	US 1962
M48 Patton	US 1952	M113A1 FSC	AU 1972	SKP/VKP	SE 1944	Light Armored Rocket Artillery		M1003 Pershing II	US 1983
M60	US 1959	Piranha IFV	CH 1974	SKPF/VKPF	SE 1956	M270 MLRS	US 1983	OTR-21 Tochka (SS-21)	SU 1975
Panzer 61	CH 1965	Pbv 301	SE 1961	Open APCs		BM-1	RU 1988	RSD-10 Pioneer (SS-20)	SU 1976
Panzer 68	CH 1971	Pbv 302	SE 1966	BTR-40	SU 1950				
SU-100	SU 1944	SPz Kurz	FR 1959	BTR-152A	SU 1950				
T-34	SU 1940	SPz Lang HS.30	CH 1960	SKP/VKP	SE 1944				
T-54	SU 1955	Stryker ICVD	US 2018	Open Armored Trucks					
T-55	SU 1958	Type 86	CN 1992	BTR-40	SU 1950				
T-62	SU 1961	Type 92	CN 1992	BTR-152A	SU 1950				
TAM	AR 1983	Type 05 IFV	CN 2005	SKP/VKP	SE 1944				
Type 59	CN 1959	Type 08 IFV	CN 2008	Open Armored Trucks					
Type 61	JP 1961	VCBI	FR 2008	BTR-40	SU 1950				
Type 69	CN 1972	Warrior	GB 1987	BTR-152A	SU 1950				
Type 74	JP 1975	YPR-765	NL 1977	SKP/VKP	SE 1956				
Type 79	CN 1984								
Type 85	CN 1988								
Type 88	CN 1988								
Type 96	CN 1997								
Type 15	CN 2018								
Vickers MBT	GB 1965								
Vijayanta	IN/GB 1965								

FCR UNITS

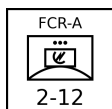
Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather. These units are presented below with their counters.

The FCR Unit Table summarizes the properties of FCR units. As well as the basic properties common to all ground units, the table gives: the FCR class; the frequency band; and the modifier to the hit roll.¹⁹

FCR-A



Towed and Mobile FCR-A Platoon: FCR-As typically use 1940s technology. They can be towed or mobile.



SCR-584: The SCR-584 is an early US towed FCR-A based on the British 3 GHz magnetron. It replaced the much larger and less accurate SCR-268 and its basic technology was used in a series of successors. The SCR-584 entered service with US and British anti-aircraft artillery units in 1944 and was commonly used to control batteries of 90 mm M2, QF 3.7-inch, 120 mm M1, and QF 5.5-inch guns. It was also supplied to the Soviet Union and used to control batteries of 85 mm KS-12, and 100 mm KS-19 guns.²⁰

SCR-784: The SCR-784 towed FCR-A is a more compact version of the SCR-584. It entered service with the US Army in 1945, replacing the SCR-584 and controlling 90 mm M2 and 120 mm M1 guns.²¹

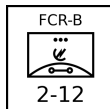
SON-4: The Soviet SON-4 (Whiff) towed FCR-A is based on the SCR-584 radars supplied by the US during WW2. It is commonly used with 85 mm KS-12 and 100 mm KS-19 batteries. It entered service in the Soviet Army in 1948 and started to be replaced by the SON-9 in 1955.²²

Blue Cedar: The Blue Cedar or "Radar, Anti-Aircraft No.3 Mk.7" towed FCR-A is a post-war development of the Mk.4 radar with post-war technology, resulting in a more compact and reliable system. It entered service with the British Army in 1952, controlling batteries of QF 3.7-inch and QF 5.25-inch guns, and served until these guns were retired in 1957. It also served with the Swiss Air Force from 1958, controlling 7.5 cm guns, until 1965, when the radars were replaced by Super Fledermaus systems and the guns by 35 mm Oerlikon GDFs. The radar was also used by the Netherlands, Sweden, Finland, and South Africa.²³

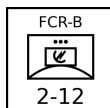
SON-9: The Soviet SON-9 (Fire Can) towed FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries. The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.²⁴

SON-30: The Soviet SON-30 (Fire Wheel) towed FCR-B was derived from the earlier SON-9. It entered service in 1957 and was used to control batteries of 130 mm KS-30 guns.²⁵

FCR-B

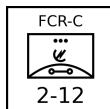


Towed and Mobile FCR-B Platoon: FCR-Bs typically use 1950s technology. They can be towed or mobile.

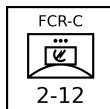


SON-50: The SON-50 (Flap Wheel) mobile FCR-B is new development and a contemporary of the RPK-2/1RL33 (Gun Dish) radar used in the ZSU-23-4. It has an integrated IFF system. It is used to control batteries of S-60 57 mm and KS-30 130 mm guns and entered service in the 1960s.²⁶

FCR-C



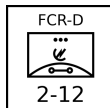
Towed and Mobile FCR-C Platoon: FCR-Cs typically use 1960s technology. They can be towed or mobile.



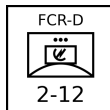
Super Fledermaus: The Contraves Super Fledermaus FCR-C can control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery. It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Israel, Japan, and South Africa. It was used by Israel in the 1967 Six Day War and afterward, controlling batteries of Bofors 40 mm L/70 guns in the defense of strategic sites. The Super Fledermaus also saw combat with the Argentine Air Force in the 1982 South Atlantic War controlling Oerlikon GDF guns in the defense of BAM Malvinas (Port Stanley Airport).²⁷

Flycatcher: The Signaal Flycatcher FCR-C can control up to six Bofors L/60 or L/70 guns or SAM launchers. Flycatcher entered service with the Royal Netherlands Air Force in 1979, serving for airfield defense and each controlling three modified Bofors 40 mm L/70 gun mounts. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987, the Thai Armed Forces from 1981, Turkey, and Venezuela.²⁸

FCR-D



Towed and Mobile FCR-D Platoon: FCR-Ds typically use 1970s technology. They can be towed or mobile.



Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles. Skyguard entered service in 1977 with the Swiss Air Force. It

FCR UNIT TABLE

Type	Size	Defense Strength		Sighting Range	Mobility	VPs		Class	FCR	
						3D/2D/D			Frequency	Modifier
Towed FCR-A	Platoon	2	12	T		6/4/2	A	LF		-1
Mobile FCR-A	Platoon	2	12	G/P		6/4/2	A	LF		-1
Towed FCR-B	Platoon	2	12	T		6/4/2	B	MF		-2
Mobile FCR-B	Platoon	2	12	G/P		6/4/2	B	MF		-2
Towed FCR-C	Platoon	2	12	T		7/5/2	C	VF		-2
Mobile FCR-C	Platoon	2	12	G/P		7/5/2	C	VF		-2
Towed FCR-D	Platoon	2	12	T		8/5/3	D	MW		-3
Mobile FCR-D	Platoon	2	12	G/P		8/5/3	D	MW		-3

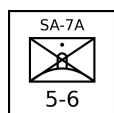
was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield) controlling Oerlikon GDF 35 mm guns.²⁹

INFANTRY SAM UNITS

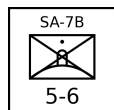
Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. These units are described below with their counters.

The Infantry SAM Unit Table summarizes the properties of infantry SAM units. In addition to the basic properties shared by all ground units, it gives: the number of ready, volley, and reload missiles; the multi-target capability; whether the unit has quick-reaction capability; whether the unit has IFF capability; whether the unit has night IR sights; and the optical lock-on number and range. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles. Infantry SAM squads may be attached to infantry, infantry weapons, infantry mortar, or infantry HQ platoons.

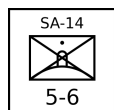
The Infantry SAM Table summarizes the properties of the actual missiles: guidance mode; launch roll; turn rate; flight time; visibility; ECCM, chaff, and flare ratings; whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability; boost phase in FPs; base speed and sustainer duration in game turns; minimum altitude (and modifier for targets at T level); hit rolls for direct and proximity hits; and damage ratings for direct and proximity hits. Most infantry SAMs do not have true proximity fuzes, and so proximity hits in the game correspond to glancing direct hits in reality.



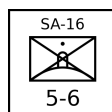
SA-7A Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker. It entered service with Soviet forces in 1968, but quickly was replaced by the improved Strela-2M (SA-7B).³⁰



SA-7B Squad: The 9K32M Strela-2M (SA-7B Grail) is similar to the Strela-2 (SA-7A Grail), but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-2M entered service with Soviet forces in 1970, replacing the Strela-2. It has seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.³¹

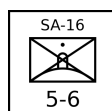


SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In par-

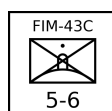


ticular, it was used by Iraqi forces in the 1991 Gulf War.³²

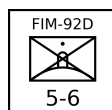
SA-16 Squad: The 9K310 Igla-1 (SA-16 Gimlet) uses as single-channel IR seeker similar to that in the SA-14 in a new missile body. The warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla-1 entered service with the Soviet Army in 1981. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters.³³



SA-18 Squad: The 9K38 Igla (SA-18 Grouse) uses a similar missile and launcher similar to the 9K310 Igla-1 (SA-16), but with a new a dual-channel IR seeker to reduce its susceptibility to flares. Again, the warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla entered service with the Soviet Army in 1983. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. It was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.³⁴



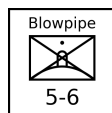
FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker. The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.³⁵



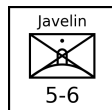
FIM-92A/B/C/D Squad: FIM-92A squad FIM-92B squad FIM-92C squad

The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide. The FIM-92A entered service with the US Army in 1981, the FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil

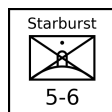
War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.³⁶



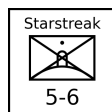
Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult. Blowpipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.³⁷



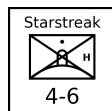
Javelin Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army.³⁸



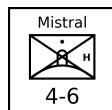
Starburst Squad: Starburst is Javelin with the radio-link replaced by a with laser link. It replaced Javelin in the British Army and Royal Marines in 1989 and was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.³⁹



Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes. Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.⁴⁰

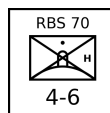


Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.⁴¹

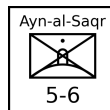


Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The

Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.⁴²



RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.⁴³



Ayn-al-Saqr Squad: The Ayn-al-Saqr ("Eye of the Hawk") missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle. It entered service with the Egyptian Army in 1984.⁴⁴

INFANTRY SAM UNIT TABLE

Name	Size	Year	Defense Strength		Sighting Range	Mobility	VPs		SAM	Missiles			Multi-Target	Quick Reaction	IFF	Night IR Sights	Lock-On		Other Names
							3D/2D/D			Ready	Volley	Reload					Optical	Range	
SA-7A	Squad	1968	5	6	U		-/-/3	SA-7A		1	1	2	-	Y	-	-	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U		-/-/4	SA-7B		1	1	2	-	Y	-	-	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U		-/-/6	SA-14		1	1	2	-	Y	-	-	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U		-/-/8	SA-16		1	1	2	-	Y	Y	-	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U		-/-/10	SA-18		1	1	2	-	Y	Y	-	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U		-/-/4	FIM-43C		1	1	2	-	Y	-	-	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U		-/-/9	FIM-92A		1	1	2	-	Y	Y	-	8	12	Stinger
FIM-92B	Squad	1986	5	6	U		-/-/10	FIM-92B		1	1	2	-	Y	Y	-	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U		-/-/10	FIM-92C		1	1	2	-	Y	Y	-	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U		-/-/10	FIM-92D		1	1	2	-	Y	Y	-	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U		-/-/6	Blowpipe		1	1	2	-	Y	Y	-	7	9	
Javelin	Squad	1984	5	6	U		-/-/8	Javelin		1	1	2	-	Y	Y	-	7	9	Javelin GL
Starburst	Squad	1989	5	6	U		-/-/8	Starburst		1	1	2	-	Y	Y	-	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U		-/-/12	Starstreak		1	1	2	-	Y	Y	-	7	12	
Starstreak LML	Squad	2000	4	6	U		-/-/14	Starstreak		3	1	0	-	Y	Y	-	7	12	
Mistral	Squad	1990	4	6	U		-/-/11	Mistral		1	1	2	-	Y	-	-	7	12	
RBS 70	Squad	1977	4	6	U		-/-/8	RBS 70		1	1	2	-	Y	Y	-	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U		-/-/4	Ayn-al-Saqr		1	1	2	-	Y	-	-	7	12	

INFANTRY SAM TABLE

Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Hit		Damage		Other Names
																	Direct	Proximity	Direct	Proximity	
SA-7A	1968	I	6	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	-	-	4	-	-	Y	-	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	-	-	4	-	-	Y	-	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	-	-	2	-	-	Y	-	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	-	-	3	-	-	Y	-	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	-	-	1	-	-	Y	-	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	-	3	0	-	-	-	-	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	-	2	0	-	-	Y	-	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	-	1	0	-	-	Y	-	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	-	1	0	-	-	Y	-	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	-	-	2	-	-	Y	-	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	-	2	0	-	-	-	-	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	-	-	5	-	-	Y	-	10	0	T	5	7	4	3	

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

The orders of battle for battalions are presented with the headquarters unit in the top row, next the organic companies, one per row, then any battalion-level assets on the next row or rows, and finally any attached units on the last line. Within each row, infantry or towed units appear on the left and their transport units on the right.

Unit Identifiers: The unit identifiers for line platoons are the platoon number, followed by the company letter or number, followed by the battalion number. If only one battalion is used, the battalion number may be omitted. Transport platoons will have a “T” prefix. For example, “1B3” means the first platoon of B company of the third battalion, and “T1B3” is its transport platoon. Companies normally have letters in Western armies and numbers in Warsaw Pact armies. Additionally, AD is used for air defense, AR for artillery, AT for anti-tank, R for reconnaissance, and S for scout, support, or security. The unit identifiers for battalion HQ units are HQ followed by the battalion number.

Unit Terminology: Within the game, units are referred to as brigades, battalions, companies/batteries, and platoons according to their size. However, some forces use other terms, but not always with the same correspondence. The Unit Terminology Table gives examples of some of these uses.

UNIT TERMINOLOGY TABLE

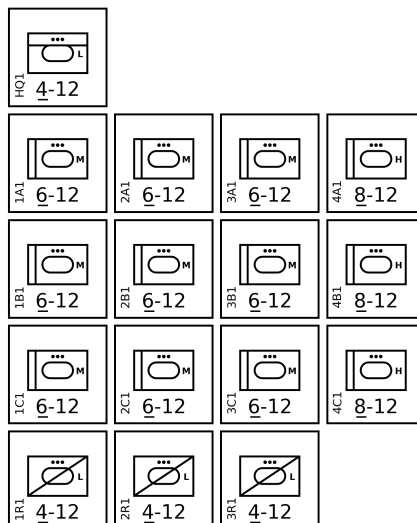
Force	Type	Brigade	Battalion	Company or Battery	Platoon
British Army	Armor	Brigade	<i>Regiment</i>	<i>Squadron</i>	<i>Troop</i>
	Artillery	—	<i>Regiment</i>	Battery	—
Soviet Army	Other	<i>Regiment</i>	Battalion	Company	Platoon
	Artillery	<i>Regiment</i>	Battalion	Battery	—
US Army	Cavalry	<i>Regiment</i>	<i>Squadron</i>	<i>Troop</i>	Platoon
	Pre-1965 Infantry	<i>Regiment</i>	Battalion	Company	Platoon
	Pre-1965 Armored	<i>Combat Command</i>	Battalion	Company	Platoon
USMC	Infantry	<i>Regiment</i>	Battalion	Company	Platoon

BRITISH ARMY (EARLY 1960S)

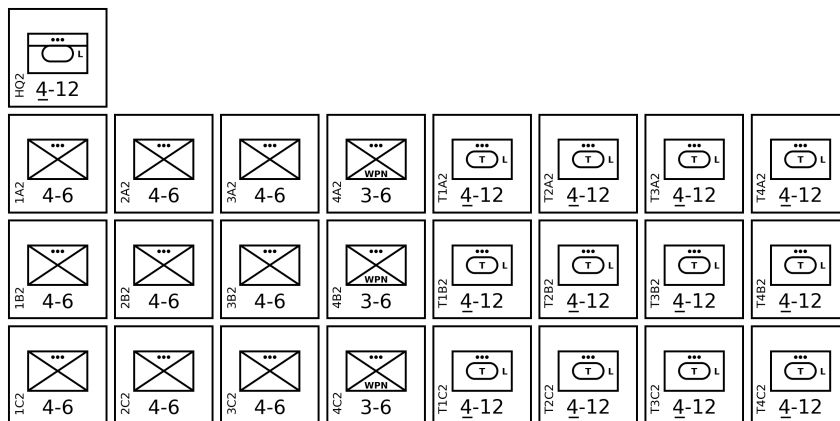
The British Army in the early 1960s was in transition. It operated both medium Centurion tanks (upgraded with the L7 105-mm gun) and heavy Conqueror tanks, and both mechanized and motorized infantry.⁴⁵

Armoured Regiment: This regiment has three squadrons A/B/C each with three tank troops with Centurion medium tanks with 105-mm guns and one tank troop with Conqueror heavy tanks with 120-mm guns. The Conqueror was originally intended to counter the Soviet IS-3 heavy tank, but after the Centurion was upgraded from the 84-mm 20-pdr gun to the 105-mm gun this need diminished. The reconnaissance squadron R has three light reconnaissance troops with Ferret armored cars.

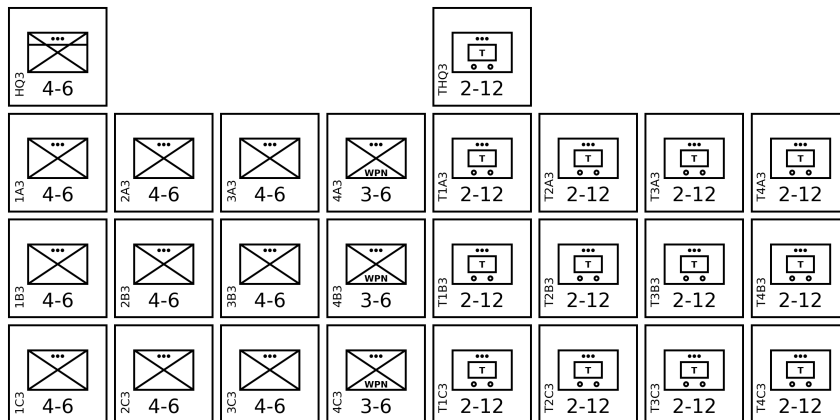
Note that in armored formations in the British Army, the terms troop, squadron, and regiment are used in place of platoon, company, and battalion.



APC Infantry Battalion: This battalion has three companies A/B/C each with three infantry platoons, one infantry weapons platoon, and corresponding light APC platoons with Saracen light APCs to provide mobility and protection. The organization parallels that of an ordinary infantry battalion, but with Saracens instead of trucks.

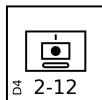
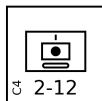
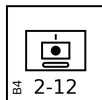
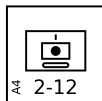
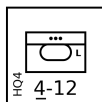


Infantry Battalion: This battalion has three companies A/B/C each with of three infantry platoons, one infantry weapons platoon, with corresponding truck platoons to provide mobility.



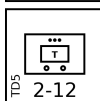
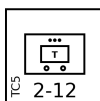
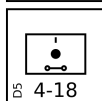
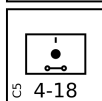
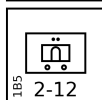
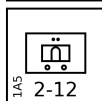
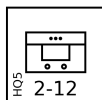
Field Artillery Regiment: This regiment has four batteries A/B/C/D each with six Cardinal 155-mm self-propelled guns.

Note that in artillery formations in the British Army, the term regiment is used in place of battalion.



Heavy Artillery Regiment: This regiment has four batteries A/B/C/D, of which A/B batteries each have two Honest John mobile missile launchers (and are each represented by section) and C/D batteries each have four towed 8-inch M1 guns with trucks.

Note that in artillery formations in the British Army, the term regiment is used in place of battalion.

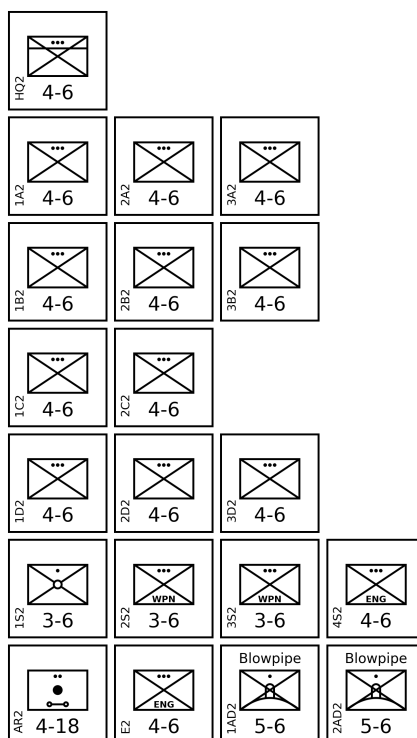


BATTLE OF GOOSE GREEN

The 1982 Battle of Goose Green in the South Atlantic War was fought between 2 Para Battalion of the British Army and 12 Infantry Regiment of the Argentine Army, both with attached elements.⁴⁶

2 Para Battalion: 2 Para consisted of: three line companies A/B/D each with three platoons of infantry; one patrol company C with two infantry platoons; and a support company S with one infantry mortar squad (as only two of the eight mortars in the mortar platoon were available), one infantry weapons platoon of sustained-fire GPMGs, one infantry weapons platoon with MILAN portable AT-GMs, and one infantry engineers platoon.

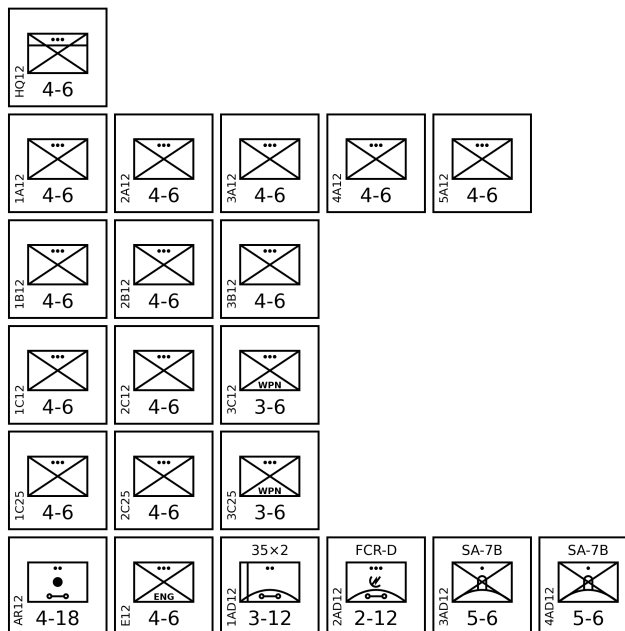
The attached units were: a towed artillery section of three 105-mm Light Guns from Alma Battery of 29 Commando Regiment, Royal Artillery; an infantry engineers platoon of 59 Independent Commando Squadron, Royal Engineers; and two squads of Blowpipes from 43 Air Defence Battery, Royal Artillery.



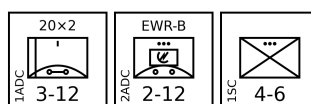
Task Force Mercedes: The Argentine Army elements at Goose Green were called *Fuerza de Taras Mercedes* (Task Force Mercedes).

The main body of the force was *Regimiento de Infantería 12* (12th Infantry Regiment), with three line companies A/B/C, and reinforced with C company of 25th Infantry Regiment.

The attached units were: a towed artillery section with three 105-mm pack howitzers from A Battery of 4th Airborne Artillery Regiment; an infantry engineers platoon from the 9th Engineering Company; a section with two Oerlikon GDF 35-mm AA guns with a Skyguard FCR from B Battery of 601st Anti-Aircraft Artillery Group; and a pair of SA-7B squads.



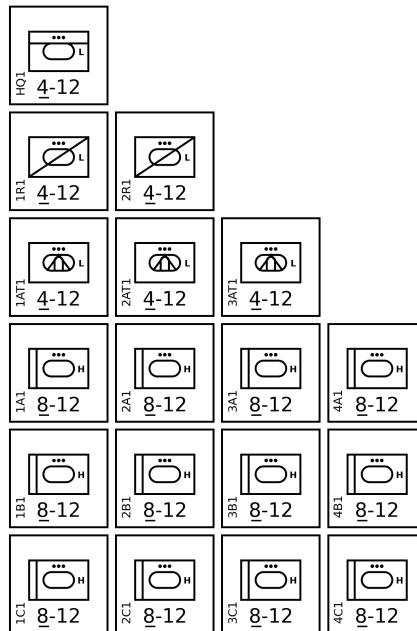
Military Airbase Condor: The Argentine Air Force named the airstrip at Goose Green as *Base Aérea Militar Cóndor* (Condor Military Air Base). The Air Force elements at the base included a battery of six 20-mm Rh-202 AA guns with an Elta EWR and a security platoon.



BRITISH ARMY (MID 1980S)

The British Army in the mid 1980s.⁴⁷

Armoured Regiment: This regiment [battalion] has three squadrons [companies] A/B/C each consisting of four tank troops [platoons] each with fourteen Chieftain heavy tanks. The reconnaissance (R) squadron [company] has two troops [platoons] each equipped with four Scorpion light tanks. The anti-tank squadron [company] has three troops [platoons] each equipped with three FV438s with the Swingfire missile.



ADDITIONAL RULES

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.⁴⁸

A. Type and Size: A ground unit will have a type, for example, “heavy tank” or “ZSU-23-4”, and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term “platoon” is more commonly used for infantry and tank units and “battery” for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground-combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.⁴⁹

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.⁵⁰

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.⁵¹ An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.⁵² A soft target does not have armored protection their personnel and/or weapons.⁵³

C. Representation: The briefing documents show the graphical representation of ground units as counters.⁵⁴ The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position. Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, tracked armored units, wheeled armored units other than armored trucks, and tracked unarmored units have unrestricted (U) mobility. Trucks, armored trucks, and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.⁵⁵

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.

When necessary, a scenario must specify whether trucks, armored trucks, and other wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.⁵⁶ A scenario may modify the mobility classes of ground units and the associated restrictions.⁵⁷

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, infantry mortar, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.⁵⁸

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, “infantry platoon with an attached FAC squad”). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3– the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.⁵⁹

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, armored truck, truck, or tracked-carrier unit of at least the same size. When necessary,

a scenario will establish the association between units and their specific transports.⁶⁰

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported towed unit is also identified (for example, “truck platoon towing an artillery battery” or “truck platoon towing a ZPU-4 battery”). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.⁶¹

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.⁶²

A towed unit may not mount or dismount during the game.

RULE 2: ANTI-AIRCRAFT ARTILLERY

A. Laser Rangefinders (LRFs): If an AAA unit has an FCR-R and an LRF, then it does not suffer a +1 modifier if the FCR is jammed or shut down. If an AAA unit has an FCR-W and an LRF, then it only suffers a +1 modifier (not a +2 modifier) if the FCR is jammed or shut down.

NOTES

1. Robin Olds

The full quotation contains a second insight: “You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb. Of course, we weren’t there to win it anyway, so it was all a waste of time.” (Sherwood, John. *Fast Movers: Jet Pilots and the Vietnam Experience*, chapter 1)

2. Ground-Combat and Transport Units

Air Strike and *Eagles of the Gulf* provide eleven types of ground-combat and transport unit: light, medium, and heavy armor, infantry, infantry FAC, infantry SAM, towed artillery, infantry HQ, armored HQ, truck, and light truck. We provide many more. Does this change outcomes? No. Effectively, all ground units are basically a protection class, defense strength, and sighting range, and the new units map to the existing ones. Does this change the experience? I think it does; the wider range of units enhances the gaming experience.

3. Infantry Weapons Platoon

An infantry weapons platoon is not strictly necessary, but it adds to the look-and-feel, and we might consider giving them longer range in ground-to-ground combat. Having a single type is probably more appropriate than having separate types for grenade launchers, recoilless rifles, ATGMs, and machine guns. One might think that the .50 cal or DShK infantry AAA unit could serve as a machine-gun weapons platoon, but machine-gun weapons platoons have low tripods for ground combat rather than high ones for AAA.

4. Infantry HQs

In the *Air Strike* scenarios HQ platoons are included at a ratio of about one to every nine normal platoons, so they seem to be battalion HQs.

5. Tanks

Protection Levels:

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8. Considering the vehicles depicted in scenarios in *Air Strike* and *Eagles of the Gulf*, it seems that typically:

- Light = protection from 14.5 mm and smaller rounds.
- Medium = protection from autocannons and guns up to about 90 mm.
- Heavy = protection against guns of 100 mm or more.

Obviously, this is a gross simplification as each vehicle has its own strengths and weaknesses, and its armor varies between its front, side, rear, and top.

To extend this to other tanks and AFVs, we use Wargames Research Group (WRG) 1950–1985 miniatures rules, which give armor classes of I to X for a range of AFVs:

CLASS I	CLASS II	CLASS III	CLASS V	CLASS IX
Sexton	Ferret	Scorpion	Vijayanta	T.72
Archer	Fox	Scimitar	Comet	T.64
VP.90	Saracen	Striker	JPz.Kan	T.10
AMX.105A	FV.432	Spartan	Panzer 68	IS.3
AMX.155F3	Abbot	Saladin	Type 61	
BTR.40	SP.70	EBR	T.34/85	CLASS IX “S”
BTR.152V	FV.180	AMX.10P	SU.100	Leopard 2
BTR.50P	AMX.105B	AMX.10RC		EPC
BTR.60P	AMX.155GCT	AMX.13	CLASS VI	T.80
ASU.57	AMX.13DCA	AMX.VC1	Leopard 1	
ZSU.57-2	AML	Luchs.	Type 74	CLASS X
SU.76	VAB	HS.30	ISU.122	Chieftain
AT-P	YP.408	IKV.91	ISU.152	Conqueror
M.7	Spz.kurz.	Staghound	M.46	Shir.1
M.42	PVB.302	M.5	M.47	Strv.103
M.44	VK.155	M.10		
M.56	BRDM	M.24	CLASS VI “s”	
M.107	BMD	M.41	Leopard.1A4	CLASS X “S”
M.110	BMP.1	Sheridan		UK.MBT.80
Otos	BTR.152K	CLASS III “s”	CLASS VI “S”	Shir.2
XR.311	BTR.50PK	FVS	AMX.32	XM.1
½ tracks	BTR.60PB	XM.723	CLASS VII	Merkava
RBV.1	PT.76	UK.MICV	Centurion 3	Leopard 3
	ZSU.23-4		T.54	
	SP.73	CLASS IV	T.55	
	SP.74	Cromwell	M.48	
	Commando	AMX.30	M.103	
	M.8	Marder		CLASS VIII
	M.18	JPz.Rak.		Centurion 13
	M.52	Gepard		T.62
	M.55	TAM		M.60
	M.59	Pz.4H		
	M.108	ASU.85		
	M.109	Sherman		
	M.113			
	M.114			
	LVTP.7			
	Lynx			
	Cougar			
	Grizzly			
	L.33			
	CLASS II “s”			
	Dutch AIFV			

The correspondence seems to be:

- Open = 4! = class I.
- Light = 4 = classes II to III.
- Medium = 6 = classes IV to VIII.
- Heavy = 8 = classes IX and X.

Since these rules are from 1979 and from unclassified sources, their reliability for the more modern vehicles is somewhat in doubt. Nevertheless, they are probably fine for older vehicles.

Heavy Tanks: The WRG 1950–1985 rules give class IX or X (i.e., heavy) protection to the Chieftain, Conqueror, EPC (Leclerc), IS-3, Leopard 2, Merkava 1, Shir 2 (Challenger 1), Strv 103, T-10, T-64, T-72, and T-80. Beyond these, we have:

- Al-Kalid/VT-1A and Type 99: These are based on the Type 90-II prototype, which in turn is based on the T-72 (heavy).
- Ariete: This weighs 54 tonnes, only slightly less than the M1A1 (heavy) which also has a crew of four, and therefore presumably has similar protection.
- Arjun: The weights 59–68 tonnes and has crew of four, similar to the M1A1 (heavy), and so presumably has similar protection.
- Challenger 2: This is derived from the Challenger 1 (heavy).
- IS-2: This has a similar weight to the IS-3 (heavy) and a similar armament (122 mm D-25), so presumably similar protection.
- K1: This is derived from the M1 (heavy).
- K2: This presumably has at least the same protection as the K1 (heavy).
- M-84, PT-91, and T-90: These are derived from the T-72 (heavy).
- Merkava 2/3/4: These are derived from the Merkava 1 (heavy).
- T-14: This presumably has at least the same protection as the T-90 (heavy).
- T-84: This is derived from the T-80 (heavy).
- Type 10: This weighs 44 tonnes in its standard configuration and has a crew of three, and so presumably has protection at least equal to earlier tanks with similar weights such as the T-72 (heavy).
- Type 90: The has a crew of three and weighs 50 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).
- Type 99: The has a crew of three and weighs 51–55 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).

Medium Tanks: The WRG 1950–1985 rules give class IV to IIX (i.e., medium) protection to the AMX-30, ASU-85, Centurion, Comet, Cromwell, ISU-122, ISU-152, Kanonen JPz, Leopard 1, M103, M4 Sherman, M46, M47, M48, M48, M60, Panzer 68, SU-100, T-34, T-54, T-55, T-62, TAM, Type 61, Type 74, and Vijayanta. Beyond these, we have:

- M26: The M46 (medium) is largely an M26 with an improved engine.
- M-50 and M-51 Sherman: These are based on the M4 (medium) and have improved armament and engines.
- Panzer 61: This is essentially an early version of the Panzer 68 (medium).
- Type 59/69/79: The Type 59 was Chinese version of the T-54A (medium). The Type 69 and 79 had improved equipment, some of which was derived from captured T-62s, but the same level of protection.
- Type 85/88/96: These have new hulls and turrets. It’s not clear whether their level of protection is significantly improved over the Type 59/69/79 (medium).
- Type 15: Wikipedia gives little information on the armor, but the weight is 36 tonnes (with the armor package), and that seems to correspond more to a medium tank.
- Vickers MBT: This is similar to the Vijayanta (medium).

Light Tanks: The WRG 1950–1985 rules give class II and III (i.e., light) protection to the AMX 10RC, AMX 13, Commando, Cougar, Fox, SpPz Luchs, M8, M24, M41, M114A2, M551, Panhard AML, Panhard EBR, PT-76, Saracen, Scimitar, and Scorpion. Beyond these, we have:

- Stryker MGS: With additional ceramic armor, this has protection against 14.5 mm but not heavier rounds, which I believe counts as III.
- Strv 74: This has a similar weight to the M41 (light).
- Type 62: Wikipedia give the hull armor as being similar to the M41 (light).
- Type 63: Wikipedia give the hull armor as being only 10 mm, which corresponds to light.
- Type 05: Based on Type 05 IFV (light).
- Type 08: Based on Type 08 IFV (light).

Open Tanks: These are uncommon, but we need an open target class for M3 half-track APCs, so it is easy to apply it here too. The WRG 1950–1985 rules give class I (i.e., open) protection to the ASU-57 and SU-76.

6. IFV Units

We might distinguish APCs and IFVs in ground-to-ground combat. Bradley IFVs made mincemeat of Iraqi tank regiments; M113 APCs would not have had the same impact.

Medium IFVs: The WRG 1950–1985 rules give class IV (i.e., medium) protection to the Marder. Beyond these, we have:

- BMP-3: Wikipedia says the BMP-3 has frontal protection from 30 mm rounds at 200 m.
- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Strf 9040/CV90: Wikipedia states that the CV90 has frontal protection against 30 mm APFSDS rounds. This is considerably more than typical light AFVs.
- LAV III (add-on-armor): Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
- M2A2/M2A3/M2A4: Wikipedia states their these M2A2 also has protection against 30 mm APDS rounds and RPGs. The M2A3 and M2A4 have visually similar armor.
- Puma: Wikipedia says the frontal arc has protection against medium-caliber rounds and shaped-charge projectiles. With additional armor, the sides gain similar protection.

Light IFVs: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the AMX-10P, BMD, BMP-1, Commando, Dutch AIFV (YPR-765), FV432 (FV432 RARDEN), HS.30 (SPz Lang), M113 (M113 FSC), SPz Kurz, UK MICV (Warrior), and XM723 (M2/M3 Bradley). Beyond these, we have:

- BMP-2: Wikipedia says the BMP-2 has similar protection to the BMP-1.
- LAV-25: Wikipedia says this only has protection against small-arms rounds.
- LAV III: Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
- Pbv 301: Wikipedia states the armor ranges from 8 to 50 mm, with upper front armor being 34 mm (20 mm at 55 degree). This seems to be light.
- Piranha IFV: Presumably similar protection to LAVs.
- Stryker ICVD: This is derived from the Stryker (light).
- Type 86: Wikipedia states this is reverse-engineered from the BMP-1.
- Type 92: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 100 meters and 25 mm rounds at 1000 meters. This seems to correspond to light armor. The weight is also 21 tonnes, which is considerable less than most medium IFVs.
- Type 05: Wikipedia states that it has protection against small-arms rounds and some protection against 12.7 mm rounds. This corresponds to light armor.
- Type 08: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 200 meter. This corresponds to light armor.
- VCBI: Wikipedia says this only has protection against 14.5 mm rounds.

There are many variants of the Piranha, and the in-service date is a guess.

7. APC Units

Heavy APCs: These are converted tanks.

- Achzarit: Wikipedia states that this based on the T-54/T-55 (medium), but has additional composite armor. Its weight at 44 tonnes is much heavier than the T-55 at 36 tonnes, despite the lack of turret, which suggests the additional armor is quite significant. Therefore, we consider it to be heavy.
- Namer: This is based on the Merkava IV (heavy).
- Nakpadon: This is an improved version of the Nagmashot and Nagmachon. Wikipedia states that its weight is around 55 tonnes, which strongly suggests its armor qualifies as heavy.
- Puma: The photos in Wikipedia show additional armor. Its weight at 50 tonnes is almost identical to the Centurion, despite the lack of turret, which suggests the additional armor is quite significant.

The in-service dates for the Nakpadon (1998) and Puma (1993) are guesses.

Medium APCs:

- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Nagmashot: Based on the Centurion (medium).
- Nagmachon: Based on the Centurion (medium).

The in-service dates for the Nagmashot (1990) and Nagmachon (2000) are guesses.

Light APCs: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BTR-50PK, BTR-60PB, Commando, Grizzly, FV432, LVTP-7, M113, M59, Saracen, Spartan, VAB, and YPR-408. Beyond these, we have:

- Bronco/Warthog: Wikipedia says it has protection against 7.62 mm rounds.
- BVS10/Viking: Wikipedia says it has protection against small-arms fire and can have additional armor to protect against 14.5 mm rounds.
- BTR-70: Wikipedia states the frontal armor is only 9 mm.
- BTR-80: Wikipedia states that this (and previous models) only have protection from small-arms fire.
- LAV II Bison: Wikipedia states that the Coyote is protected against small-arms rounds, and the Bison seems to be similar.
- LVTP-5: Wikipedia states that the maximum armor is 16 mm.
- M75: Wikipedia states the maximum armor is 38 mm. This is more than the thickness of some light tanks, but it is almost vertical.
- OT-64 SKOT: Wikipedia states the maximum armor is 13 mm.
- OT-810: Wikipedia states the maximum armor is 14.5 mm.
- Piranha APC: Presumably similar protection to LAVs.
- M1126 Stryker: Wikipedia states it has some protection against 14.5 mm rounds from the front and all-round protection against small-arms fire.
- TPz Fuchs: Wikipedia states that it is protected against small-arms rounds.
- Type 63: Wikipedia states that it is protected against small-arms rounds.
- Type 77: Wikipedia states that it is based on the PT-76 (light), and its weight of 15–5 tonnes matches other light APCs.
- Type 89: Wikipedia states that it is protected against small-arms rounds.
- Type 08: Based on Type 08 IFV (light).

There are many variants of the Piranha, and the in-service date is a guess.

Open APCs: The WRG 1950–1985 rules give class I (i.e., open) protection to the AT-P, BTR-50P, BTR-60P, and Half-tracks (M3).

8. Armored Truck Platoon

Mobility:

- The BTR-40 is based on the GAZ-63 truck.
- The BTR-152 is based on the ZIS-151 or ZIL-157 truck.
- The SPK/VPK are based on SAAB/Volvo flat-bed trucks.
- The Saxon has only four wheels, which makes me suspect it has limited mobility.

Light Armored Trucks: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BTR-152K. Beyond this, we have:

- Saxon: Wikipedia states that it is protected against small-arms rounds.

Open APCs: The WRG 1950–1985 rules give class I (i.e., open) protection to the BTR-40 and BTR-152. Beyond these, we have:

- SKP/VKP/SKPF/VKPF: Wikipedia states the armor is 8 to 20 mm, which would be light if the vehicle were not open.

Sources:

- BTR-40: Wikipedia.
- BTR-152: Wikipedia.
- Saxon: Wikipedia.
- SKP/VKP/SKPF/VKPF: Wikipedia

9. Armored Anti-Tank Missile Vehicles

Anti-tank units are mainly chrome. However, in ground combat, we might give them more effectiveness against vehicular or towed targets and less against infantry targets.

Medium Anti-Tank Missile Vehicles: The WRG 1950–1985 rules give class IV (i.e., medium) protection to the Raketen JPz. Beyond this, we have:

- Jaguar: This is Raketen JPz (medium) upgraded with HOT missiles.

Light Anti-Tank Missile Vehicles: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BRDM and Striker. Beyond these, we have:

- Fennek MRAT: Based on the Fennek (light).
- FV438: This is derived from the FV432 APC (light).

- M150 and M901: These are derived from the M113 APC (light).
- M1134 Stryker ATGM: These are derived from the Stryker APC (light).
- VAB HOT: This is derived from the VAB APC (light).
- YP-408 PRAT: This is derived from the YPR-408 APC (light).
- YPR-765 PRAT: This is derived from the YPR-765 APC (light).

10. Armored Reconnaissance Vehicles

Again, reconnaissance vehicles are mainly chrome. However, in ground combat we might give them reduced effectiveness.

Light Reconnaissance Vehicles: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BRDM-2, Ferret, Lynx (M113½ Lynx), and M114. Beyond these, we have:

- LAV II Coyote: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.
- Fennek: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.

Light Reconnaissance Vehicles: The WRG 1950–1985 rules do not give any reconnaissance vehicles with class I (i.e., open) protection. However, we have:

- M20: Open version of the M8 (light) with no turret and armed only with a .50-cal machine gun.

11. Armored Mortars

For the protection class, we have:

- FV432: based on FV432 (light).
- M106/M125/M1064: based on M113 (light).
- M1129 Stryker MCV: based on Stryker (light).

We might consider these as open.

12. Armored Artillery

Light Armored Artillery: The WRG 1950–1985 rules give class II (i.e., light) protection to the Abbot, AMC-155GCT (AMX-30 AuF1), L33, M52, M108, M109, SP.73 (2S3), SP.74 (2S1), and VK.155 (Bkan 1). Beyond these, we have:

- 2S19 Msta-S: Wikipedia states this all-round protection from 15 mm of steel armor. This corresponds to light armor.
- AS-90: Wikipedia states this has up to 17 mm of steel armor. This corresponds to light armor.
- Panzerhaubitze 2000: Wikipedia states this all-round protection from 14.5 mm of steel armor. This corresponds to light armor.

Open Armored Artillery: The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX-105A, M7, and M44. Beyond these, we have:

- The M-50 has frontal and side armor, like the M7 and unlike for example the M107 and M108.
- The M40 also has frontal and side armor. The rear door is probably armored too, but is open when the gun is used.

13. Armored Rocket Artillery

Light Armored Rocket Artillery: For the protection class, we have:

- M270: I can find no definite information on the armor protection.
- BM-1: According to Wikipedia, the T-72 hull was chosen to give adequate protection close to the front lines. However, it is not clear whether the rocket launcher has adequate protection.

14. Tracked Artillery

To count as a hard target, both the crew and weapons must be protected by armor.

- The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX.155F3 (AMX-13 F3), M107, and M110. However, the only crew member protected seems to be the driver. The other crew members travel and operate the gun without protection.
- Similar considerations apply to the 2S4, 2S7 (although some of the crew travel under armor), M40, M41, and Mk F3. In all of these, the gun mount is open.

15. Mobile Artillery

To count as a hard target, both the crew and weapons must be protected by armor.

- CAESAR: Some models have lightly-armored cabins, but the truck as a whole does not seem to be protected.
- Archer: The crew and engine compartments is protected against small-arms rounds, but the gun does not seem to be protected. Unlike the CAESAR, the gun is operated by the crew from the protected cabin. This is close to reaching the level of protection of light armored artillery.

16. Mobile Rocket Artillery

To count as a hard target, both the crew and weapons must be protected by armor.

- M142 HIMARS: The cabin has some protection, but the rest of the truck seems to be unprotected.
- BM-30: The cabin appears to have some protection, but the rest of the truck seems to be unprotected.
- Earlier Soviet vehicles appear to be completely unarmored.

17. Mobile Rocket Artillery

To count as a hard target, both the crew and weapons must be protected by armor, which is not the case here.

- M752 Lance: This seems to be based on the unarmored M548.
- AMX-30 Pluton: The vehicle is based on an AMX-30 hull and seems to be armored, but the missile is exposed.

Tracked missile artillery is mainly included to serve as a target.

18. Mobile Missile Artillery Batteries

Missile artillery is mainly included to serve as a target.

Regarding the protection class, none of these have complete protection against small arms.

19. FCR Units

Bands: Radar frequencies are often referred to by the corresponding NATO (post 1978) and IEEE band designations.

Frequency (GHz)	NATO Band	Frequency (GHz)	IEEE Band
2–3	E	1.55–3.9	S
3–4	F	3.9–6.2	C
4–6	G	6.2–10.9	X
6–8	H	10.9–20.0	Ku
8–10	I	20.0–36.0	Ka
10–20	J		
20–40	K		

Frequencies: The table gives a summary of the frequencies of various FCRs along with the NATO and IEEE bands. Some radar systems use different individual radars for different roles. The roles are R for range, S for search, and T for tracking. The sources of the data are given in the entries for the individual radars.

Radar	Year	Role	Frequency (GHz)	NATO Band	IEEE Band	MTI	IFF	Notes
SCR-584	1944	S/T	2.7–2.8	E	S	N		
SCR-784	1945	S/T	2.7–2.8	E	S	N		
SON-4	1945	S/T	2.70–2.86	E	S	N		
Blue Cedar	1952	S/T	3.00–3.12	F	S	N		
SON-9	1955	S/T	2.70–2.86	E	S	N		
SON-30		S/T	—	E	S	N	Y	
SON-50		S/T	9.25–9.70	I	X	N	Y	
Super Fledermaus	1965	S/T	8.6–9.6	I	X	Y	N	MTI from 1970
Flycatcher	1979	S	9.2	I	X	Y	Y	MTI from 1992?
		T	9.2	I	X	—	—	
		T	26.5–30	K	Ka	—	—	From 1992?
Skyguard	1977	S/T	8.6–9.5	I	X		Y	
M163/M167	1967	R	9.2–9.5	I	X	—	N	
Panhard M3 DCA	1975	S/R	—	E	S	Y	N	
ZSU 23-4	1965	S/T	14.6–15.6	J	Ku	N	Y	IFF from 1977
Sinai 23	1992	S/R	—	E	S	Y	N	
Tunguska	1984	S	—	E	S	Y	Y	
		T	—	J	X	—	—	
Gepard	1976	S	—	E	S	Y	Y	
		T	—	J	Ku	—	—	
M51	1953	T	—	I	X	N		
AMX-13 DCA	1969							
AMX-30SA	1979							
BOFI-R								

20. SCR-584

I classify this as FCR-A as the SON-9 is derived from it.

Wikipedia gives the frequency as 2.7–2.8 GHz (NATO E-band) and the search and tracking ranges for bomber-sized targets as 64 km and 29 km.

Werrell mentions its use with the 90 mm and 3.7-inch guns against V-1s in 1944.

Sources:

- Wikipedia on the SCR-584, 90 mm M1/M2 gun, 120 mm M1 gun, and QF 3.7-inch AA gun.
- Werrell, K.P. (2005, Air University Press, 2nd edition): “Archie to SAM”.
- RadarTutorial
- MobileRadar on the SCR-784

21. SCR-784

I classify this as FCR-A it is derived from the SCR-584.

The earliest manuals I can find are from 1945, so I suspect that is its in-service date.

Sources:

- Wikipedia on the SCR-784 and SCR-584.
- RadarTutorial on the SCR-784 and SCR-584
- MobileRadar on the SCR-584
- RadioNerds on the SCR-784.

22. SON-4

I classify this as FCR-A it is derived from the SCR-584.

RadarTutorial states that the frequency is 2.70–2.86 GHz.

RadarTutorial states that it is a tracking radar, and was typically used with a P-8 (Knife Rest A) or P-10 (Knife Rest B) search radar.

Sources:

- RadarTutorial on the SON-4.

23. Blue Cedar

I'm not sure whether this is an FCR-A or FCR-B.

The frequency is 3.00–3.12 GHz (NATO F-band and IEEE S-band).

The Swiss FCRs seem to have controlled 75 mm Flab Kan 38 L39 guns.

Sources:

- Wikipedia on the Blue Cedar.
- RadarTutorial on the Blue Cedar.
- Wikipedia on the Swiss Fliegerabwehrtruppen.

24. SON-9

Wikipedia gives the frequency as 2.7–2.8 GHz (NATO E-band). RadarTutorial gives it as 2.7–2.86 GHz.

Wikipedia states this was used with the 57, 85, 100, and 130 mm guns and was widely deployed in the Vietnam War. In *The Speed of Heat* the 57 mm S-60 and 85 mm KS-12 are often combined with an FCR-A, presumably the SON-9.

Sources:

- Wikipedia on the SON-9.
- RadarTutorial

25. SON-30

This looks very similar to the SON-9.

RadarTutorial gives the frequency as S band.

The photo on RadarTutorial shows an integrated IFF system.

Wikipedia and RadarTutorial both state it was used for 130 mm guns, but do not mention others.

RadarTutorial states that it had replaced the SON-4 and SON-9 by 1957, but the SON-9 only entered service in 1955. I suspect that it entered service in 1957.

Foss (p. 185) states that it was used for 130 mm guns.

Sources:

- Wikipedia on the SON-30.
- RadarTutorial on the SON-30.
- Foss, Christopher F., 1974 (London : Allan): “Artillery of the World”.

26. SON-50

This is truck-mounted.

RadarTutorial gives the frequency as 9.25–9.7 GHz (X band).

The photo on RadarTutorial shows an integrated IFF system.

RadarTutorial states it was used for 57 and 130 mm guns. Wikipedia states it was used for 57 mm guns.

RadarTutorial states that it is mounted on a URAL-375. Wikipedia then states that this was produced from 1961 to 1993. The in-service data can be no earlier than 1961. I guess it entered service in the 1960s; by the 1970s I think the focus had moved to SAMs.

Chant (p. 253) states that the SON-50 is believed to be derived from the Gun Dish radar on the ZSU-23-4 and has a TV back-up. I note, however, that the frequencies are very different: Flap Wheel is E band and Gun Dish is J band.

Sources:

- Wikipedia on the SON-50 and Ural-375
- RadarTutorial on the SON-50.
- Chant, Christopher, 1989 (Brassey's Defence Publishers): “Air Defence Systems and Weapons World AAA and SAM Systems in the 1990s”.

27. Super Fledermaus

Wikipedia states there were two versions: the Feuerleitgerät (Flt Gt) 63 (from 1965) and the Feuerleitgerät 69 (from 1970). The latter had “background clutter suppression,” which probably is equivalent to MTI.

Wikipedia states that it can control *two* Oerlikon GDF mounts, hence the restriction to controlling sections and not batteries.

Wikipedia also states these the prototype of the Gepard used a Flt Gt 63. Since the FCR-W radar of the Gepard is equivalent to a +2 modifier and has VF frequency, it corresponds to an FCR-C. We assume the Super Fledermaus does too.

The Wikipedia page on the Super Fledermaus gives the frequency as 8.6–9.6 GHz (old NATO X-band), but also states that its search function operates in the E/F bands (2–4 GHz) and tracking function in the J band (10–20 GHz).

The service history is from Wikipedia and Singh (for Israeli use).

Sources:

- Wikipedia on the Super Fledermaus and its use with the Oerlikon GDF.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

28. Flycatcher

Flycatcher is similar to the Cheetah radar, so presumably FCR-C.

According to Wikipedia, the search radar is NATO I/J band and the tracking radar is NATO K band. RadarTutorial states the search frequency is 9.2 GHz (NATO I band) and the tracking frequency is either I or K band. Van Elk states the tracking radar is an independent pulse-Doppler 26.5–31 GHz (old-NATO Ka band) system, but this seems to be referring to an upgrade in 1992. He also mentions that the radar is used in the Goalkeeper CIWS, which Wikipedia states searches in I band and tracks in both I and K bands. I conclude:

- Search is 9.2 GHz (I band)
- Tracking is 9.2 GHz (I band) or 26.5–31 GHz (K band)

Van Elk states that it has IFF and directed a maximum of 3 cannons or missile launchers.

Van Elk states that it succeeded the HSA L4/5 radar.

Sources:

- Wikipedia on the Flycatcher.
- van Elk, Bert (2020): *Terug naar de toekomst: De Flycatcher*.
- RadarTutorial

29. Skyguard

RadarTutorial states that the frequency is 8.6–9.5 GHz (NATO I-band or IEEE X-band). The description of the system suggests that both search and tracking radars use the same frequency.

Sources:

- Wikipedia on the Skyguard's use with the Oerlikon GDF.
- RadarTutorial

30. SA-7A Squad

The launcher and SAM values are from *Air Strike* with significant modifications:

- Zaloga gives an in-service date of 1968.
- *Air Strike* gives this a seeker type of I, but that does seem appropriate for an uncooled PbS seeker. Instead, I use a seeker type of E, by analogy with the uncooled seeker in the AIM-9B, and works at about 2 microns.
- *Air Strike* gives a turn rate of HT, but the control capability seems to be the same as in the SA-7B and SA-15. I will adopt BT.

- *Air Strike* gives this a speed of 8. Zaloga gives a lifetime of 8 seconds and an average speed of 1550 km/h. This corresponds to a range of 5.2 km or about 10 hexes.

31. SA-7B Squad

The launcher and SAM values are from *Air Strike* with significant modifications:

- *Air Strike* gives this a seeker type of I. That seems about right. The seeker was PbS and cooled electrically, like the AIM-9E which also has a type I seeker.
- *Air Strike* gives a turn rate of BT, which I adopt.
- *Air Strike* gives this a speed of 10. Zaloga states that it has a slightly more powerful rocket motor than the SA-7A, but the mean speed is the same at 1550 km/h. I adopt a speed of 10, like that of the SA-7A.

32. SA-14 Squad

The launcher and SAM values are from *Air Strike* with significant modifications:

- The seeker is cryogenically cooled at works at about 4 microns (Zaloga). *Air Strike* gives this a type A seeker, but that does seem appropriate; the first type A seekers in Soviet AAMs did not appear until 1985 in the AA-8C. Also, type A seekers have all-aspect capability Zaloga says that the seeker has similar performance to the RIM-43C, but only limited all-aspect capability. A type M seeker seems to be more appropriate.
- *Air Strike* gives a turn rate of ET, but the control capability seems to be the same as in the SA-7A and SA-7B. I will adopt BT.
- *Air Strike* gives this a speed of 12. Zaloga gives a lifetime of 8 seconds and an average speed of 1440 km/h (as it is slightly heavier than the SA-7). This corresponds to a range of 4.8 km or about 9 hexes.
- The warhead is the same as on the SA-7A/7B.

33. SA-16 Squad

The launcher and SAM values are from Malcolm Pipes with significant modifications:

- The warhead of the SA-16/18 is 1.17 kg, identical to the 1.17 kg of the SA-7 and similar to the 1.1 kg of the SA-14. Only later Iglas have heavier warheads. Therefore, I have reduced the damage ratings to match the SA-7.
- Carl Smeaton comments that the warhead explodes any remaining rocket fuel. I increase the damage rating by one if the target is reached in the first half of the FPs.
- I have reduced the lock-on range of the SA-16 and increased the flare rating to match those of the SA-14.
- Furthermore, I have given both the SA-14 and SA-16 the same speed and turn rate. Zaloga states that the engagement range increased by about 1 km with respect to the SA-7/14, so I am increasing the speed by 2 to 12. I will give both a turn rate of BT/2, matching Stinger.

34. SA-18 Squad

The launcher and SAM values are from Malcolm Pipes with some modifications:

- The warhead of the SA-16/18 is 1.17 kg, identical to the 1.17 kg of the SA-7 and similar to the 1.1 kg of the SA-14. Only later Iglas had heavier warheads. Therefore, I have reduced the damage ratings to match the SA-7.
- Carl Smeaton comments that the warhead explodes any remaining rocket fuel. I increase the damage rating by one if the target is reached in the first half of the FPs.
- I have given both the SA-14 and SA-16 the same speed and turn rate. Zaloga states that the engagement range increased by about 1 km with respect to the SA-7/14, so I am increasing the speed by 2 to 12. I will give both a turn rate of BT/2, matching Stinger.

35. FIM-43C Squad

The launcher and SAM values are from *Air Strike*.

36. FIM-92A/B/C/D Squad

The launcher and SAM values for the A model are from *Air Strike*. Those for the B/C/D models are from Malcolm Pipes compendium.

- I have increased the flare susceptibility of the FIM-92A from 2 to 3. It did not seem to have significant protection.
- Wikipedia gives an effective range of 5 miles, which is equivalent to a speed of 15. I have left the speed at 14, as that is close enough.
- Wikipedia gives a peak speed of 1930 mp and a self-destruct time of 17 seconds.

- *Air Strike* has the visibility as 2, but looking at photos I don't see much differ

37. Blowpipe Squad

The launcher and SAM values are from *Air Strike*.

38. Javelin Squad

The launcher and SAM values for Javelin are from *Air Strike*.

Javelin is also called "Javelin GL".

I have seen mention of a Javelin LML; did anyone use it or was it just a prototype?

39. Starburst Squad

The launcher and SAM values for Starburst are just those for Javelin but with the OG guidance replaced by LG guidance.

Starburst was originally called "Javelin S15".

40. Starstreak Squad

The launcher and SAM values are from Malcolm Pipes. The real darts do not have proximity fuzes, but the missile in the game can obtain a proximity hit. Perhaps a proximity hit here means a hit with only one dart?

41. Starstreak LML Squad

The SAM values are from Malcolm Pipes. I have guessed the launcher values are the same as for shoulder-launched version, just that the launcher has three ready missiles. I don't know if the launcher has reloads. Is the launcher really portable or does it have to be transported by a vehicle?

42. Mistral Squad

The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

43. RBS 70 Squad

For the RBS 70: The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

44. Ayn-al-Saqr Squad

The launcher and SAM values are taken to be equal to those of the SA-7B.

45. British Army Units (Early 1960s)

The British Army formations from early 1960s are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

46. Goose Green

The primary source for the orders of battle at Goose Green is Mark Adkin's *The Battle for Goose Green: A Battle is Fought to be Won* (Leo Cooper, 1992).

47. British Army Units (Mid 1980s)

The British Army formations from mid 1980s are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

48. Rules

Much of this section should be included in the 3A rules, but it's here for the moment.

49. Squads and Sections as Damaged Platoons

So for example sections/squads have 2D/1D damage resilience and +1/+2 modifiers to AAA attacks. Similar modifiers should be applied to ground-to-ground attacks.

50. Damage Resilience in *Air Strike*

There is a form of damage resilience in *Air Strike*, in that units whose counters have white text are only able to take 1D of damage before being killed (see the last paragraph of rule 28). Such units include infantry SAMs, armored CCUs, and many vehicle SAMs. *The Speed of Heat* also mentions units being able to take only 2D damage, but it doesn't have any relevant units since it attaches SAM teams to infantry platoons rather than having them as separate (see below) and doesn't have a separate FAC unit. In both cases, the idea seems to have been incompletely implemented in the rules. I'm generalizing and homogenizing the idea. This is important as we need sections for mobile AAA units with two vehicles and for the Oerlikon GDF (as Skyguard can only control two guns and not a full battery). To a large degree, a section is equivalent to a platoon that has taken 1D and a squad is equivalent to a platoon that has taken 2D, so the mechanics of the game are really not impacted.

51. Hard Targets

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8 in *Air Strike*.

52. Open Targets

The BTR-50P APC and the M3 half-tracked APC are examples of open targets. In the 1956 and 1967 Arab-Israeli Wars where both sides used M3 half-tracks. If an M3 is attacked by an air-burst or napalm bomb, the crew and passengers will suffer more or less as if they were in the open.

53. Soft Targets

Examples of soft targets include infantry, towed artillery, unarmored vehicles, and armored vehicles that do not fully protect their personnel and/or weapons such as the M-107 and M-110 tracked artillery vehicles.

Nothing like the M-107 or M-110 is included in *Air Strike*, but again I would assert that it is tracked mainly for mobility. Effectively, these are towed artillery installed in an exposed position on a tracked hull and have almost no protection for most of the crew.

Most tracked SAM vehicles are classified as hard targets in *Air Strike* and *Eagles of the Gulf*, despite many of them having completely exposed missiles. If an air-burst bomb explodes above a SAM-4 launcher, it is not going to shrug it off. I think many SAM vehicles are tracked mainly for *mobility* and perhaps for crew protection, but in terms of a mission-kill on their weapon systems, they are soft targets. I've not changed anything yet, but we need to think about this.

54. Representation

The representation is based on NATO symbols, including those for infantry, armor, anti-armor, air defense, guns, missiles, mortar, tube artillery, and rocket artillery. One addition is that a rectangle denotes an unarmored vehicle such as a truck or tracked carrier. A truck or mobile unit has two wheel symbols in the lower position, whereas a tracked carrier has a tracked symbol. The mortar symbol is modified to be simply an open circle, for compactness.

All vehicular units have either an armored or unarmored vehicle symbol, all infantry units have an infantry symbol, and towed units have none of these.

The letter T in the central position indicates a transport capacity (e.g., APCs, IFVs, cargo trucks, and tracked cargo carriers). Words and letters in the lower position indicate variants on a basic type: the letters H, M, L, and O on an armored unit indicate heavy, medium, light, and open armored protection; the word WPN or FAC on an infantry unit indicates a weapons or FAC unit; and the letter H on an infantry SAM unit indicates a heavy version of the launcher (e.g., the LML version of Starstreak).

55. Mobility

The inspiration for this rule was this photograph of the Mitla Pass from 1967, which shows trucks destroyed on and just off a road. Also, in the Falklands War roads could be traversed by Landrovers and AMLs, but off-road terrain was impassable to everything except the CVR(T)s and Bv 202s despite being open.

56. Truck Mobility

For example, a scenario set in the open desert of southern Iraq might specify that trucks have good mobility, but one set in the Falkland Islands might specify that they have poor mobility. Similarly, a scenario might specify that cargo trucks have poor mobility, but mobile artillery units have good mobility. Finally, a scenario might not assign an explicit transport unit to a towed artillery battery, but might state that it has been transported by helicopter and so can be placed as if it had a transport unit with good mobility.

57. Modifying Mobility Classes

One common modification is to explicitly indicate the hexes occupied by each ground unit; in this case this rule is largely superfluous (although if we allow ground units to move, then the restrictions can still be relevant).

Panhard AMLs are wheeled armored vehicles and so normally have unrestricted mobility. Nevertheless, in the Falklands War they were restricted to roads and trails, and this corresponds to poor mobility. A scenario could fix this problem by explicitly giving them poor mobility.

58. Composite Platoons

This rule comes from *The Speed of Heat*. It doesn't have counters for infantry SAMs, so in scenarios V-11 and V-22 it says treat infantry platoons as if they have an infantry SAM squad. This rule helps with stacking (since attached squads do not count for stacking) and with keeping your precious FACs and SAM teams anonymous. It also works nicely with the transport rule, since attached squads do not need their own transport but can ride with the platoon.

59. Damage to Attached Squads

The roll of 4- seems about right. The probability of killing an attached squad is 40% for 1D and 64% for 2D. Those numbers are fairly good approximations to

1/3 and 2/3. If the roll was 3-, they would be 30% and 51%, and that seems a little low.

Are the damage rolls secret?

60. Transportation

The idea of transporting units and cargo comes from the "Thunder and Fire" scenario from *Eagles of the Gulf*. The rules here simply generalize it slightly, incorporate the idea of mobility, and add rules for mounting and dismounting.

61. Mobility

Being placed according to the mobility of its transport allows a tracked transport to place a towed unit in a forest hex.

62. Mounting and Dismounting Infantry

The rules for mounting and dismounting are completely new. I'm very open to suggestions about this.